



Figure 1: Sencha Touch



Figure 2: Apache Cordova

- **HTML5 device APIs:** A lot of native device APIs can be added to the packager API surface, which make it compatible with all Web browsers.
- **Native packaging:** This inbuilt feature of Sencha Touch can now package applications for mobile platforms such as Android and iPhone.

Apache Cordova/PhoneGap

Apache Cordova (formerly known as PhoneGap) is an open source mobile application development framework originally created by Nitobi. After Nitobi was bought out by Adobe Systems, the company redesigned PhoneGap and released the software as Apache Cordova, which allows developers to use Web technologies like HTML5, CSS3 and JavaScript for cross-platform development. Cordova facilitates wrapping up of CSS, HTML and JavaScript code, depending on the platform of the device. The core of Apache Cordova applications makes use of CSS3 and HTML5 for rendering, and uses JavaScript for logic. In order to access hardware like the accelerometer, camera and the GPS, HTML5 is being used. Cordova can also be extended with native plugins, which add more functionalities via JavaScript, making the communication direct between the native layer and the HTML 5 page.

Apache Cordova supports the development of various mobile app platforms like iOS, Bada, BlackBerry, Firefox, Windows Phone, Android, WebOS, Symbian, Tizen and Ubuntu Touch.

The unique features of Apache Cordova are:

- Cordova is regarded as a JavaScript API acting as a wrapper for native code and is consistent across devices. It is regarded as an application container with a Web view, which covers the entire screen of the device.
- Cordova is bundled with predeveloped plugins to support access to the phone's camera, GPS and file system.
- Cordova installs just like native applications. It will produce an IPA file for iOS, APK file for Android and XAP file for Windows Phone.



Figure 3: Appcelerator Titanium



Figure 4: Qt

Appcelerator Titanium

Appcelerator Titanium is an open source mobile app development framework for iOS, Android and Windows Phone from a single JavaScript code base. Appcelerator provides JavaScript API and native platform-specific features across devices. In addition to

providing a development framework, it offers value add-ons including app analytics, Backend-as-a-Service (BaaS) and others. This tool has an edge in providing code reusability while creating native iOS and Android apps.

Its features are:

- Provides JavaScript based SDK with other 5000 APIs for iOS, Android, Windows, BlackBerry and HTML5.
- Developers can make use of their Web development skills to build rich and fully native mobile apps for almost any mobile platform.
- Reuses 60-90 per cent of code when supporting multiple platforms.
- Has a cross-platform APU for accessing native UI components such as navigation bars, menus and dialogue boxes and native device functionality including the file system, network geolocation, accelerometer and maps.

Qt

Qt is an important cross-platform mobile application framework mostly used for developing application software that can run on various software and hardware platforms with little or no change in the underlying code base. Qt is mainly used for developing various application software with or without GUI interfaces like command line tools, mobile apps, etc.

Qt chiefly uses C++ with signal and slot extensions, which makes it easy when handling events and helps in both developing the GUI and the server applications. Qt supports many compilers like G++ and even Microsoft Visual Studio. It has a declarative scripting language called QML, which facilitates using JavaScript being integrated with any sort of development. With activated support for the development of various apps and software for desktops and mobile platforms, Qt declares the UI layouts using QML-JSON like declarative languages. It supports development in various